

The Japan Factor Puzzle: All of the Crash Risk, None of the Premium

Code and data: [<https://github.com/GT-HLab/quant-research/tree/master>].

Over the past 35 years, one dollar invested in the European momentum factor grew to roughly \$28. The same strategy in Japan returned \$1.10. Momentum, arguably the most robust anomaly in asset pricing, has not produced a measurable premium in Japan in any decade on record. This piece documents how deep that failure runs, rules out the easy explanations, and sets up the question: what is Japan telling us about how factor premia actually work?

Why this matters

Most global factor allocations treat momentum as a single, universal premium harvested across regions. The data say otherwise: over 1990–2026, the long-short momentum factor ranges from a t-statistic of 4.80 in Europe to 0.59 in Japan. Japan is not a mild underperformer within a working global strategy. It sits at one end of a spread that runs from 10.6% in Europe to 1.4% in Japan, the regional choice mattering more than the factor choice, and understanding *why* is worth more than the allocation fix.

Data and method

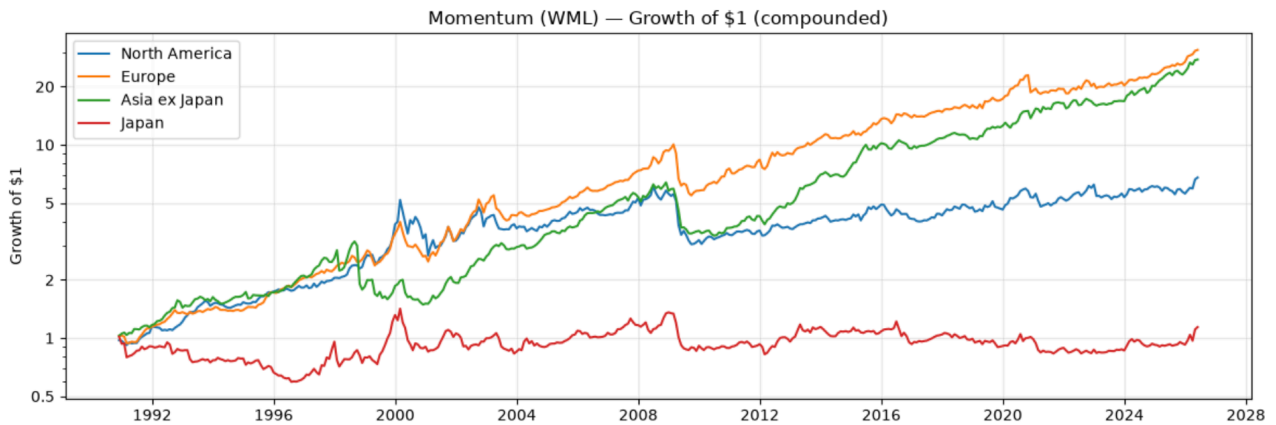
All results use the Kenneth French developed-markets data library: monthly long-short momentum factor returns (winners-minus-losers, 12-1 formation) and total market returns for North America, Europe, Japan, and Asia Pacific ex Japan, November 1990 through mid-2026. Returns are raw factor returns, long-short portfolios are self-financing, so no risk-free adjustment applies. No transaction costs, no implementation constraints. One caveat up front: the regional series begin in late 1990, which excludes the strong early decades of US momentum documented on longer samples; full-sample comparisons between North America and Europe should be read with that in mind.

The dispersion is the setup

Region	Momentum Ann. Mean (%)	Momentum Ann. Vol (%)	Momentum Sharpe	Momentum t-stat	Market Max DD (%)	Market Max DD Date	Momentum Max DD (%)	Momentum Max DD Date
North America	6.63	15.71	0.42	2.52	-51.07	2009-02-28	-49.97	2009-09-30
Europe	10.58	13.13	0.81	4.80	-58.97	2009-02-28	-45.59	2009-09-30
Asia ex Japan	10.48	14.31	0.73	4.37	-60.05	2009-02-28	-52.81	2000-11-30
Japan	1.43	14.48	0.10	0.59	-58.10	2003-03-31	-41.96	2012-02-29

[Table 1 : Summary - Momentum Factor Return Statistics per Region]

Europe delivered 10.6% annualized at a 0.81 Sharpe ($t = 4.80$). Asia ex Japan is close behind at 10.5% ($t = 4.37$). North America managed 6.6% ($t = 2.52$), significant by the classical threshold. Japan: 1.4% annualized, $t = 0.59$, statistically indistinguishable from zero across 35 years.



[Graph 1 : growth of \$1 invested in Momentum Factor, log scale]

Compounded, the dispersion is brutal. A dollar in European or Asian ex-Japan momentum grew to roughly \$28. In North America, about \$6. In Japan, \$1.10, a flat red line across three and a half decades.

Not one bad decade: the failure is stable

Period	Region	Ann. Mean (%)	Ann. Vol (%)	Sharpe	t-stat
1990–1999	North America	15.48	11.29	1.37	4.15
	Europe	14.09	9.74	1.45	4.38
	Asia ex Japan	8.99	19.53	0.46	1.39
	Japan	4.54	17.50	0.26	0.79
2000–2012	North America	1.89	21.18	0.09	0.32
	Europe	8.93	17.38	0.51	1.85
	Asia ex Japan	8.45	13.51	0.63	2.25
	Japan	-1.27	15.16	-0.08	-0.30
2013–2026	North America	5.19	11.38	0.46	1.67
	Europe	9.77	10.04	0.97	3.56
	Asia ex Japan	13.45	10.37	1.30	4.75
	Japan	1.91	11.23	0.17	0.62

[Table 2 : Momentum Factor Return statistics - sub-period table]

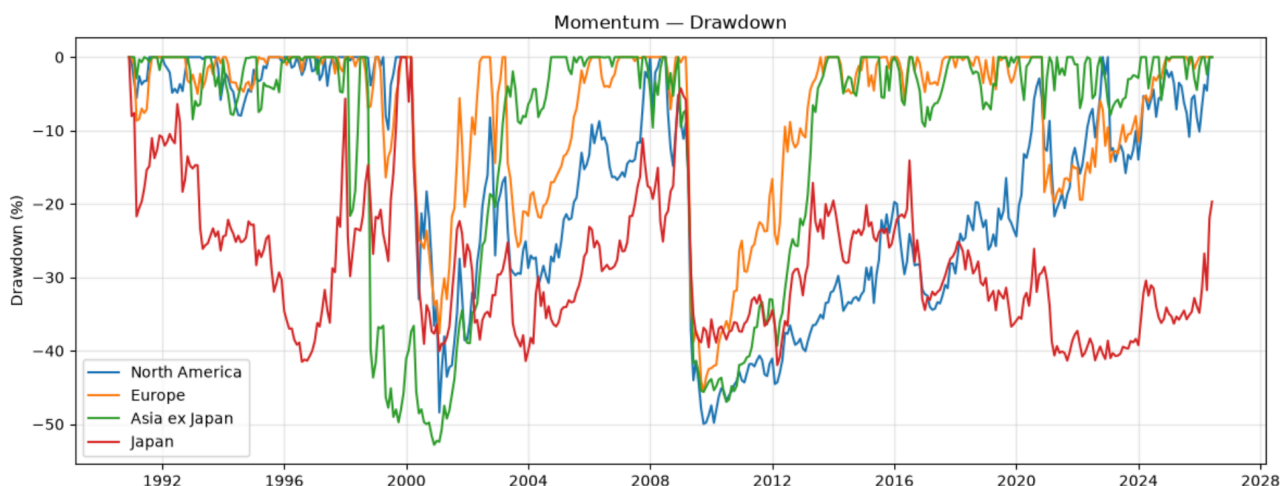
Splitting the sample into 1990–1999, 2000–2012 and 2013–2026 shows how differently the regions behave, and how uniquely Japan fails.

North America decayed: $t = 4.15$ in the 1990s, then 0.32, then 1.67. Full-sample significance carried entirely by the first decade. Asia ex Japan inverted: insignificant in the 1990s, strengthening ever since ($1.39 \rightarrow 2.25 \rightarrow 4.75$). Europe alone is significant in all three eras. These are three different regimes: decay, migration, persistence.

Japan fits none of them. Its momentum premium is insignificant in every sub-period: t-statistics of 0.79, -0.30 and 0.62. Not a decaying premium, not a late arrival, not one catastrophic decade dragging down an otherwise healthy series. Through the bubble unwind, through the global momentum crash of 2009, the Japanese result never moves. Whatever is happening in Japan is persistent, and it has survived three distinct global regimes.

Japan crashes with everyone and never recovers

Momentum's worst losses do not arrive with the crash. They arrive with the recovery. Regional markets bottomed in February 2009, but momentum's maximum drawdowns in North America and Europe landed in September 2009, six months into the rebound (Table 1). Asia ex Japan shows the same pattern in a different cycle, with its worst momentum loss in November 2000, in the reversal that followed the Asian financial crisis. The standard explanation is that momentum is implicitly short a crash-recovery option: After a market collapse, the short leg fills with crushed high-beta names. When the market rebounds, those names rip and the factor bleeds out.



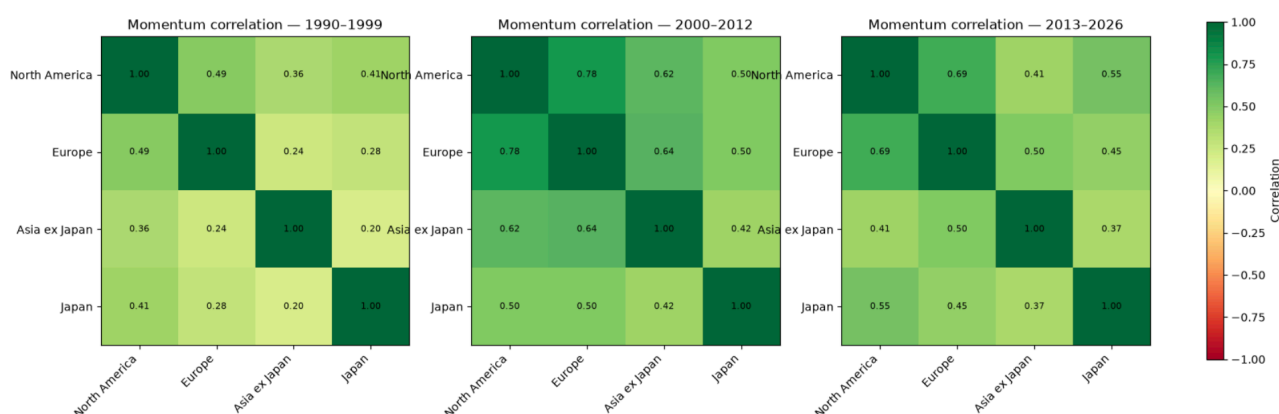
[Graph 2 : Momentum Factor Return underwater chart]

Japan participates fully in this mechanism. Its momentum drawdowns are comparable in depth to everyone else's (-42% among regional losses of -42% to -53%). The difference is what comes after. European momentum recovered its 2009 losses in roughly four years. North America needed close to nine. Japan's momentum factor hit its maximum drawdown in February 2012 and has never returned to its prior peak; it sits roughly 20% below it today. While the Japanese *market* recovered, Japanese *momentum* did not. All of the crash risk, none of the repair.

Stating the puzzle precisely

Two claims need separating.

The first is a statistical claim: *Japan's true momentum premium is zero*. The data cannot establish this. A t-statistic of 0.59 is a failure to reject zero, not a demonstration of it. With 14.5% annualized volatility, thirty-five years cannot distinguish a small premium from no premium at all



[Graph 3 : Regional Momentum Factor Correlation Matrix by sub-period]

The second claim is the puzzle: *Japanese momentum carries the risk profile of the global factor while paying nothing measurable*. If Japanese momentum were mere noise, a market where the factor simply doesn't operate, its correlation to other regions would sit near zero. It doesn't over the 35 years period: 0.47 to North America, 0.42 to Europe, moderate, but far from independent, and rising over time as the regions integrated (Japan-NA moves from 0.41 in the 1990s to 0.55 today). Japan is not running its own uncorrelated experiment. It sits inside the same global momentum regime, absorbs the same crashes and delivers a premium indistinguishable from zero across 35 years.

That combination is what makes Japan interesting rather than merely disappointing. A market where momentum was noise would be a curiosity. A market that is plugged into global momentum's risk machinery, yet where the premium never materializes, is a stress test for every theory of why momentum pays at all.

What could explain it?

Candidate explanations are well rehearsed: a historically mean-reverting market in which trends fail to persist, cross-shareholding structures that dampen price continuation, a different investor base and analyst dynamic. The data in this piece cannot distinguish between them. What the data do not support is the convenient answers: Japan's momentum failure does not line up with its weak index, is not confined to one bad era, and is not statistical noise disconnected from the global factor.

There is one classic resolution in the literature, and it points directly at where this series goes next. Asness (2011) argued that Japan is "the exception that proves the rule": on his data, Japanese momentum's weakness was offset by an unusually strong Japanese value premium, and the two combined worked in Japan roughly as well as anywhere. It is an elegant answer, but it was written on data ending fifteen years ago. Whether the offset survived is an open question.

A practitioner's note

The results above use academic decile-sort factors. A cross-sectional risk-model (Barra-style construction) would decompose the same phenomenon differently: pure factor returns are industry- and exposure-neutralized, so part of what a sorted portfolio calls "momentum" is reallocated to industries, beta and size. The regional dispersion documented here is a statement about investable sorted portfolios. The pure-factor version of the Japan question is a separate exercise.

Limitations

Sample begins November 1990. Long-short academic factors ignore shorting costs, borrow availability and financing. Decile sorts are not investable portfolios. One factor definition (12-1, monthly rebalance) and four regional aggregates. And the central caveat bears repeating: the Japanese result is a failure to reject zero, not a demonstration of zero.

Next

Momentum needs trends to persist. Value needs prices to revert. If Japan is the market that punishes the first, the classic argument says it should be the market that rewards the second. Part 2: does the Japanese value premium still hold, and did the governance-reform era change the answer?

Personal research; views are my own.